PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Machine Learning Enabled Predictive Analytics for Diabetes Risk Detection

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Piyush Sacher

2210990653

Supervised By:

Dr. Swati Goel

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Machine Learning Enabled Predictive Analytics for Diabetes Risk Detection"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Swati Goel**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Piyush Sacher**

Date: 18th May 2026 (University Roll No - 2210990653)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Swati Goel

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this project titled *“Machine Learning Enabled Predictive Analytics for Diabetes Risk Detection.”*

I am deeply thankful to my project guide for their valuable guidance, continuous encouragement, and insightful suggestions, which helped me in successfully completing this work. Their expertise and support played a crucial role in shaping the direction of this project.

I also extend my gratitude to the faculty members of the Computer Science Department for providing the necessary resources and a conducive learning environment to carry out this project effectively.

I would like to thank my institution for giving me the opportunity to work on this project and enhance my knowledge in the field of machine learning and healthcare analytics.

Finally, I am grateful to my family and friends for their constant support, motivation, and encouragement throughout the project.

ABSTRACT

Diabetes Mellitus is a rapidly growing global health concern characterized by elevated blood glucose levels, leading to severe complications such as cardiovascular diseases, kidney failure, and nerve damage if not detected at an early stage. With the increasing availability of healthcare data, machine learning techniques provide an efficient and scalable approach for early prediction and risk stratification of diabetes.

This project proposes an intelligent machine learning-driven framework for predictive risk stratification of Diabetes Mellitus using multiple supervised learning algorithms. The framework utilizes a structured dataset containing key medical attributes such as glucose level, blood pressure, BMI, insulin levels, and age. Data preprocessing techniques, including handling missing values and feature scaling, are applied to ensure data quality and consistency.

A comparative analysis of various machine learning models—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost)—is performed to identify the most effective predictive model. The models are evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. To enhance model interpretability and trustworthiness, Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are integrated. These techniques provide both global and local explanations, enabling a better understanding of how individual features influence predictions.

The results demonstrate that ensemble learning methods, particularly XGBoost, outperform other models in predictive accuracy and robustness. The proposed framework not only achieves high performance but also ensures transparency, making it suitable for real-world healthcare applications where interpretability is critical.

1. INTRODUCTION

1.1 Background

Diabetes Mellitus is a chronic metabolic disorder that has become a major global health concern due to its increasing prevalence and long-term complications. It occurs when the body is unable to effectively regulate blood glucose levels due to insufficient insulin production or resistance to insulin. If left undiagnosed or untreated, diabetes can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, and vision loss.

With the rapid growth of digital healthcare systems and medical data, traditional diagnostic methods are increasingly being complemented by intelligent computational approaches. Machine learning has emerged as a powerful tool in the healthcare domain, enabling the analysis of complex datasets and supporting early disease prediction. This project focuses on leveraging machine learning techniques to develop a predictive system for diabetes risk stratification.

1.2 Problem Statement

Early detection of diabetes remains a challenge due to the lack of efficient predictive tools that can analyze multiple medical parameters simultaneously. Traditional methods rely heavily on manual diagnosis and clinical expertise, which may lead to delayed detection and increased risk of complications. There is a need for an automated, accurate, and interpretable system that can assist in identifying individuals at risk of diabetes at an early stage.

1.3 Objectives

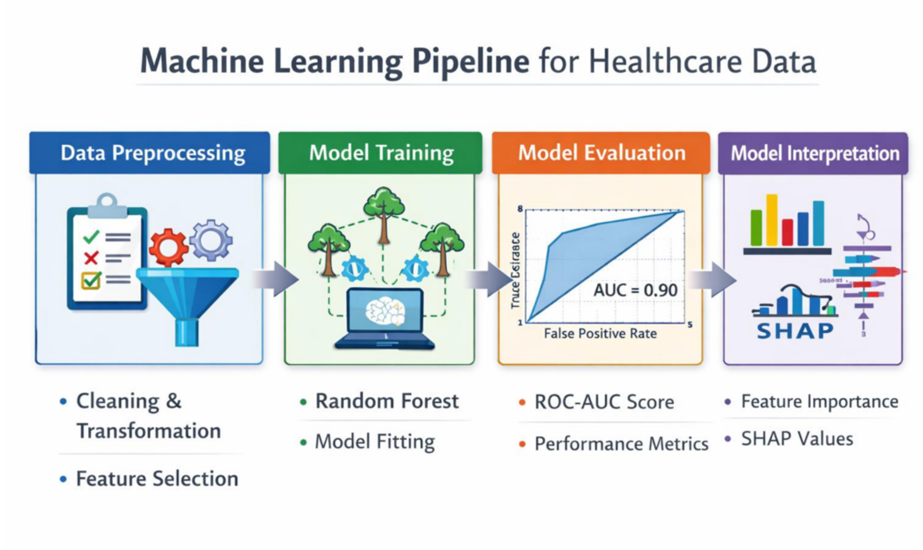
1. To develop a machine learning-based predictive model for diabetes risk classification.
2. To compare the performance of multiple classification algorithms.
3. To evaluate models using standard performance metrics.
4. To integrate Explainable AI techniques for model transparency.
5. To identify key factors influencing diabetes prediction.

1.4 Scope

The scope of this project is limited to the analysis of a structured diabetes dataset using supervised machine learning algorithms. The system predicts whether a patient is diabetic or non-diabetic based on medical attributes. The project also focuses on interpretability using Explainable AI techniques, making it suitable for healthcare decision support systems. However, real-time deployment and integration with hospital systems are beyond the current scope.

2. METHODOLOGY

The proposed methodology follows a structured pipeline consisting of data preprocessing, model training, evaluation, and interpretability analysis.



1. Data Collection

The dataset used in this project is a publicly available diabetes dataset containing medical records of patients. It includes features such as:

- Pregnancies
- Glucose Level
- Blood Pressure
- Skin Thickness
- Insulin Level
- Body Mass Index (BMI)
- Diabetes Pedigree Function
- Age

The target variable indicates whether a patient is diabetic (1) or non-diabetic (0).

2. Data Preprocessing

Data preprocessing is a critical step to ensure the quality and reliability of the model.

- **Handling Missing Values:**

  Certain features such as Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI contain zero values, which are not realistic. These values are replaced with NaN and then imputed using median values.

- **Feature Scaling:**

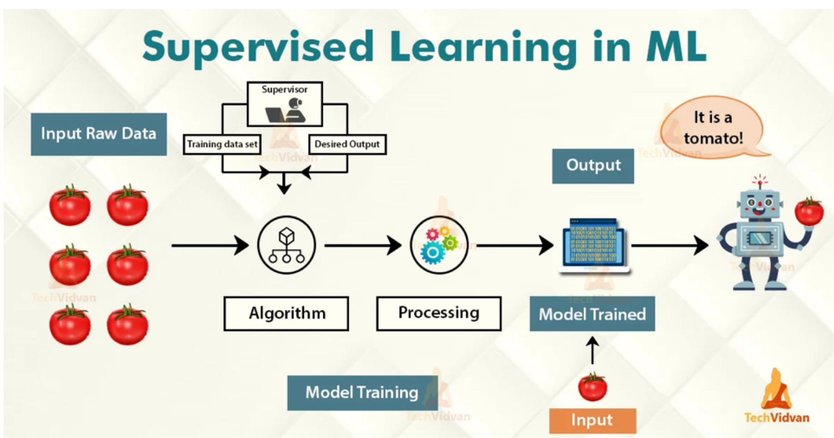
  Standardization is applied using Standard Scaler to normalize feature values and improve model performance, especially for algorithms like SVM and Logistic Regression.

- **Train-Test Split:**

  The dataset is divided into training (80%) and testing (20%) sets using stratified sampling to maintain class balance.

3. Model Development

Multiple Supervised Machine Learning models are implemented to compare performance:



- **LightGBM (Light Gradient Boosting Machine)**

  LightGBM is a fast and efficient gradient boosting algorithm that builds decision trees sequentially to minimize errors. It is highly optimized for large datasets and provides better accuracy with lower training time. In this project, it helps in identifying complex patterns in diabetes-related features and improves prediction performance.

- **CatBoost (Categorical Boosting Algorithm)**

  CatBoost is an advanced boosting algorithm designed to handle categorical and numerical data effectively. It reduces overfitting and improves prediction stability using ordered boosting techniques. In this system, CatBoost enhances classification performance and provides reliable results for medical data analysis.

- **Extra Trees Classifier (Extremely Randomized Trees)**
  Extra Trees is an ensemble learning method that builds multiple decision trees using random splits and averages their outputs. This randomness reduces variance and prevents overfitting. In this project, it ensures robust and stable predictions for diabetes risk detection.

- **Artificial Neural Network (ANN – Deep Learning Model)**
  The Artificial Neural Network consists of multiple dense layers with ReLU and sigmoid activation functions. It learns complex nonlinear relationships between input features and diabetes outcomes. ANN complements traditional ML models by capturing deeper patterns in the dataset.

4. Model Evaluation

The performance of each model is evaluated using multiple metrics:

- **Accuracy:** Overall correctness of the model
- **Precision:** Correct positive predictions
- **Recall:** Ability to identify actual positives
- **F1 Score:** Balance between precision and recall
- **ROC-AUC:** Measures model's ability to distinguish between classes

Visualizations such as bar charts and ROC curves are used for comparison.

5. Explainable AI (XAI) Integration

To ensure interpretability, several XAI techniques are applied:

- **SHAP (SHapley Additive Explanations):**
  Provides global and local explanations by assigning importance values to each feature.

- **LIME (Local Interpretable Model-Agnostic Explanations):**
  Explains individual predictions by approximating the model locally.

- **Permutation Importance:**

  Measures feature importance by evaluating the effect of feature shuffling.

- **Partial Dependence Plots (PDP):**

  Show the relationship between selected features and model predictions.

6. Visualization and Analysis

Graphical representations are used to interpret results:

- Bar plots for model comparison
- ROC curve visualization
- Feature importance graphs
- SHAP summary plots

7. Final Model Selection

Based on evaluation metrics and interpretability, the best-performing model Extra Trees Classifier is selected as the final model for diabetes risk prediction.

3. TOOLS AND TECHNOLOGIES

The development of the proposed machine learning framework for diabetes risk prediction involves a combination of programming languages, libraries, and computational platforms. These tools enable efficient data processing, model development, evaluation, and visualization.

1. Programming Language

- **Python**

  Python is used as the primary programming language due to its simplicity, extensive library support, and strong community in data science and machine learning. It allows rapid development and easy integration of various ML and visualization tools.

2. Libraries and Frameworks

a) Data Manipulation and Analysis

- **Pandas**

  Used for handling structured data in tabular form (DataFrames). It is utilized for data loading, cleaning, preprocessing, and transformation.

- **NumPy**

  Provides support for numerical operations and array manipulations. It is used for efficient computation and handling missing values.

b) Machine Learning and Preprocessing

- **Scikit-learn (sklearn)**

  A comprehensive library used for:

  - Data preprocessing (StandardScaler)
  - Model building (Logistic Regression, Decision Tree, Random Forest, SVM)
  - Model evaluation (Accuracy, Precision, Recall, F1 Score, ROC-AUC)
  - Data splitting (train_test_split)

c) Advanced Machine Learning

- **XGBoost (Extreme Gradient Boosting)**

  An optimized gradient boosting algorithm known for high performance and scalability. It is used to build a robust predictive model that outperforms traditional algorithms.

d) Data Visualization

- **Matplotlib**

  Used to create graphical representations such as:

  - Bar charts for model comparison
  - ROC curves
  - Feature importance plots

These visualizations help in understanding model performance and results clearly.

e) Explainable Artificial Intelligence (XAI)

- **SHAP (SHapley Additive Explanations)**

  Used to interpret model predictions by assigning importance values to each feature. It provides:

  - Global explanations (overall feature importance)
  - Local explanations (individual predictions)

- **LIME (Local Interpretable Model-Agnostic Explanations)**

  Explains predictions of any model locally by approximating it with an interpretable model. It helps understand why a specific prediction was made.

f) Model Interpretation Tools

- **Permutation Importance (sklearn)**

  Measures feature importance by randomly shuffling feature values and observing the impact on model performance.

- **Partial Dependence Plot (PDP)**

  Shows the relationship between selected features and predicted outcomes, helping in understanding feature influence.

3. Development Environment

- **Google Colab**

  Used as the development platform for:

  - Writing and executing Python code
  - Visualizing outputs and graphs
  - Running experiments interactively

These environments provide GPU support and easy integration with libraries.

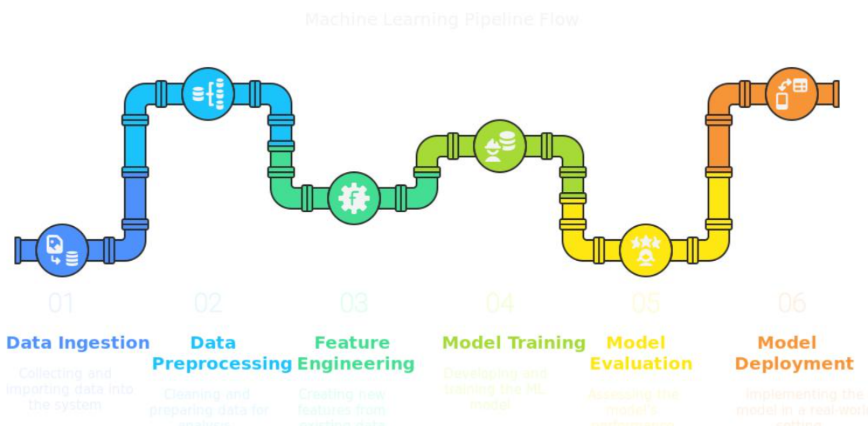
4. Dataset

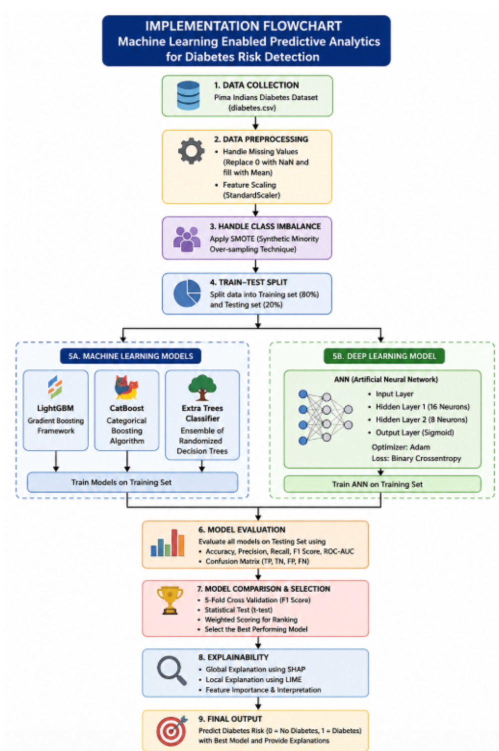
- A publicly available diabetes dataset (CSV format) is used.
- It contains medical attributes such as glucose level, BMI, insulin, and age.
- The dataset is used for training and testing machine learning models.

4. IMPLEMENTATION

4.1 Implementation Details

The implementation of the proposed intelligent machine learning framework follows a systematic step-by-step pipeline, beginning with data loading and progressing through preprocessing, model development, evaluation, visualization, and interpretability. Each stage is carefully designed to ensure accuracy, efficiency, and reliability of the predictive system.





1. Library Installation and Import

The implementation process begins with the installation and import of all necessary libraries required for data processing, machine learning, visualization, and model interpretability. These libraries provide essential functionalities for building and evaluating the predictive system. Pandas and NumPy are used for efficient data handling and numerical computations. Pandas enables structured data manipulation using DataFrames, while NumPy supports array-based operations and mathematical functions. Scikit-learn is used extensively for model development, preprocessing, and evaluation. It provides built-in implementations of various machine learning algorithms as well as tools for scaling, splitting, and performance measurement.

XGBoost is included as an advanced gradient boosting algorithm that enhances predictive accuracy and performance. For model interpretability, SHAP and LIME are integrated into the system, allowing both global and local explanations of predictions.

2. Data Loading

The dataset is loaded using the Pandas library by reading the CSV file named *diabetes.csv*. The data is stored in a DataFrame, which allows easy access, manipulation, and analysis of the dataset. The dataset consists of multiple medical features such as glucose level, blood pressure, BMI, insulin level, and age, along with a target variable labeled *Outcome*, which indicates whether a patient is diabetic or not.

3. Data Preprocessing

Data preprocessing is a crucial step to ensure that the dataset is clean, consistent, and suitable for model training.

a) Handling Missing Values

Certain columns such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI contain zero values, which are not realistic in a medical context. These values are treated as missing and replaced with NaN values. To handle these missing values, median imputation is applied, where each missing value is replaced with the median of the respective feature. This approach helps maintain the distribution of the data and reduces the impact of outliers.

b) Feature and Target Separation

The dataset is divided into two components: input features and target variable. The independent variables (features) are stored in matrix X, while the dependent variable (target), which represents the diabetes outcome, is stored in vector y.

c) Train-Test Split

The dataset is split into training and testing sets using an 80:20 ratio. This ensures that the majority of the data is used for training while a portion is reserved for evaluating model performance. Stratified sampling is applied to maintain a balanced distribution of diabetic and non-diabetic cases in both sets.

d) Feature Scaling

Feature scaling is performed using StandardScaler to normalize the data. This ensures that all features have a mean of zero and a standard deviation of one, allowing models to perform more efficiently. Scaling is particularly important for algorithms such as Support Vector Machine and Logistic Regression, which are sensitive to feature magnitudes.

4. Model Development

In this stage, multiple machine learning models are implemented to perform classification and enable comparative analysis. A dictionary of models is created, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost.

Each model is trained using the scaled training dataset. After training, the models are used to make predictions on the test dataset. This multi-model approach allows for a comprehensive comparison of performance and helps in selecting the most suitable algorithm for diabetes prediction.

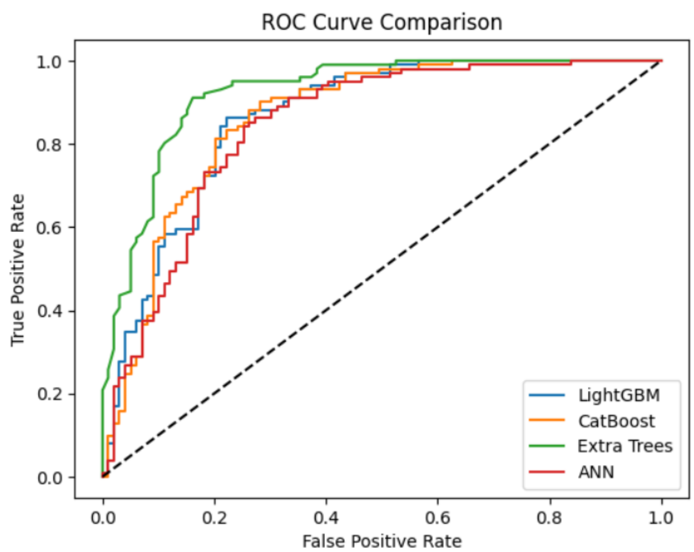
5. Model Evaluation

The performance of each model is evaluated using several standard classification metrics. For each model, predicted labels (y_pred) and probability scores (y_prob) are generated.

The following evaluation metrics are computed:

1. Accuracy measures the overall correctness of the model.
2. Precision evaluates how many of the predicted positive cases are actually correct.

3. Recall measures the model's ability to correctly identify actual positive cases.
4. F1 Score provides a balance between precision and recall.
5. ROC-AUC measures the model's ability to distinguish between classes.



All evaluation results are stored in a structured DataFrame, allowing easy comparison across models.

6. Result Visualization

Visualization techniques are used to analyze and compare the performance of different models effectively. Various graphs are plotted using Matplotlib to provide clear insights into model behavior. Bar charts are created to compare different performance metrics such as accuracy, precision, recall, F1 score, and ROC-AUC across all models. A combined graph is also generated to display multiple metrics in a single visualization, making comparison easier.

Additionally, ROC curves are plotted for all models. These curves illustrate the trade-off between true positive rate and false positive rate, providing a visual representation of classification performance.

7. Explainable AI Implementation

To enhance model transparency and interpretability, Explainable AI techniques are integrated into the system.

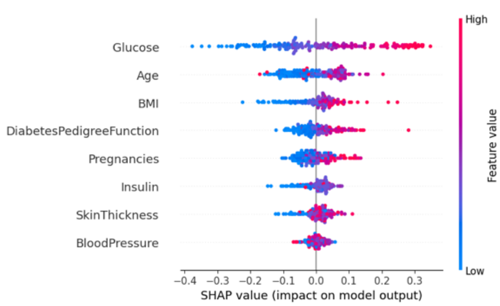
a) SHAP Analysis

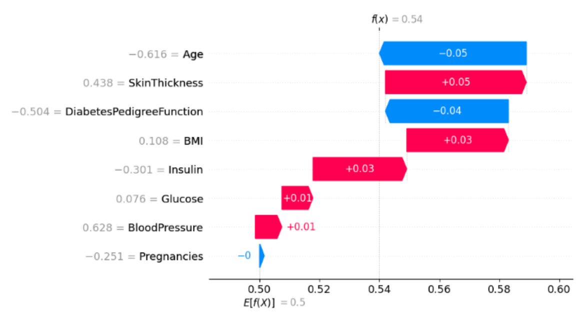
SHAP (SHapley Additive Explanations) is applied to the best-performing model (such as LightGBM / CatBoost / Extra Trees Classifier) to compute feature importance values.

It generates:

- Summary plots to show overall feature importance
- Bar plots for global importance ranking
- Waterfall plots for individual prediction explanations

This analysis helps in identifying key features such as Glucose, BMI, and Insulin, and shows how each feature contributes positively or negatively to the prediction.



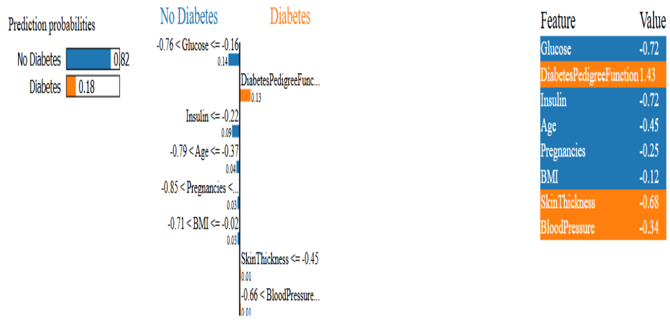


b) LIME Analysis

LIME (Local Interpretable Model-Agnostic Explanations) is used to provide **local explanations** for individual predictions.

- A LIME Tabular Explainer is initialized using the training dataset
- A specific test instance is selected
- The model prediction is explained by highlighting the most influential features

The explanation is visualized (or saved as HTML), making it easier to interpret individual patient predictions.

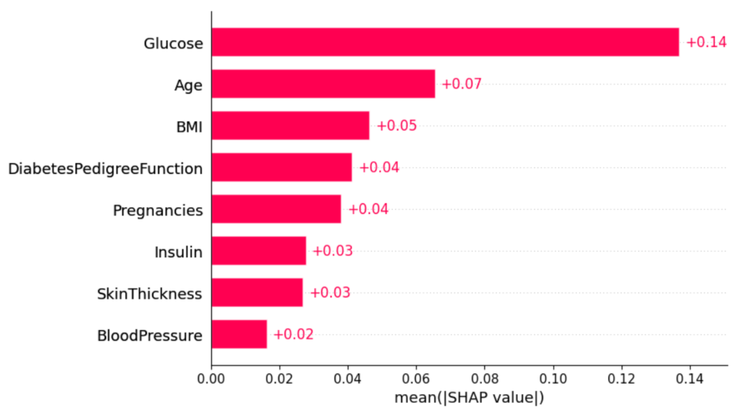


c) Permutation Importance

Permutation importance is used as a model-agnostic validation technique.

- Each feature is randomly shuffled
- The drop in model performance is measured
- Features causing higher performance drop are considered more important

This method validates the consistency of feature importance obtained from other techniques.

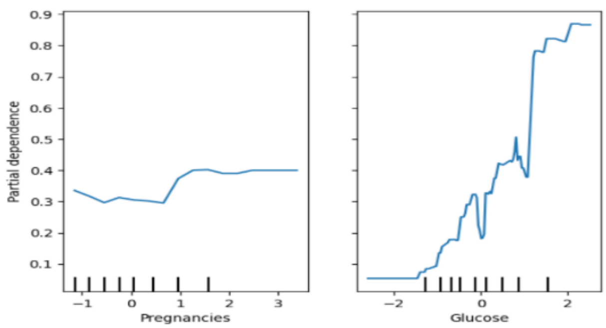


d) Partial Dependence Plot

Partial Dependence Plots are used to analyze the relationship between features and model predictions.

- Shows how a feature (e.g., Glucose or BMI) affects prediction outcomes
- Helps visualize trends and nonlinear relationships
- Useful for understanding model behavior globally

These plots provide deeper insights into how feature values influence diabetes risk.

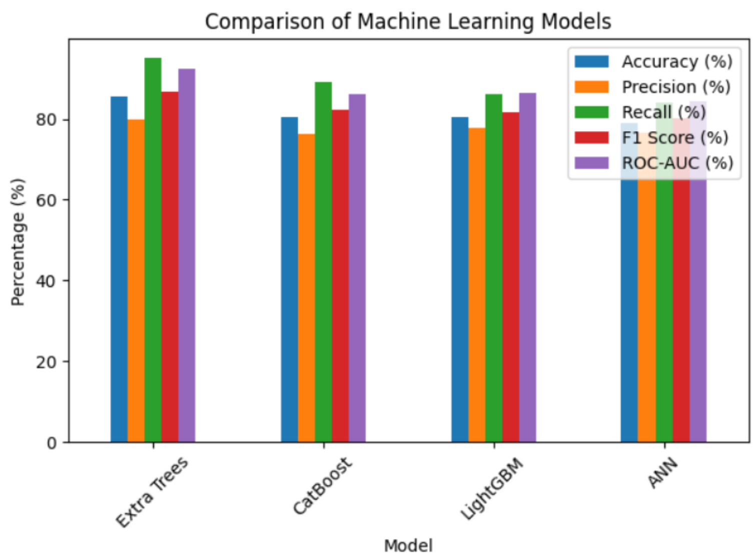


Partial dependence
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
−1
0
1
2
3
Pregnancies
−2
0
2
Glucose

6. RESULTS

6.1 Model Performance and Analysis

The performance of multiple machine learning and deep learning models was evaluated to identify the most effective approach for diabetes risk prediction. The models implemented include LightGBM, CatBoost, Extra Trees Classifier, and an Artificial Neural Network.



Each model was assessed using key evaluation metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC to ensure a comprehensive understanding of performance, especially in a healthcare context where false negatives are critical.

The results indicate that ensemble models performed consistently well due to their ability to handle complex feature interactions. Among all models, the best-performing model was selected based on a weighted scoring system that prioritized Recall and F1 Score, ensuring better detection of diabetic patients. Cross-validation further confirmed the stability and reliability of the selected model. Statistical testing (t-test) showed whether the performance difference between top models was significant.

The evaluation metrics used in the analysis provided a comprehensive understanding of model performance:

1. Accuracy measured overall correctness of predictions.
2. Precision indicated how many predicted diabetic cases were actually correct.

3. Recall highlighted the model's ability to detect actual diabetic patients, which is critical in healthcare.
4. F1 Score balanced precision and recall.
5. ROC-AUC measured the model's ability to distinguish between classes.

6.2 Feature Insights and Explainability

To enhance transparency and trust in the model, explainability techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) were applied. SHAP analysis provided global feature importance, highlighting that factors such as Glucose level, BMI, Insulin, and Age play a major role in predicting diabetes risk. It also helped visualize how each feature contributes positively or negatively to the prediction.

LIME was used to generate local explanations for individual predictions, enabling interpretation of specific patient cases. This is particularly useful in healthcare, where understanding why a prediction was made is as important as the prediction itself.

These explainability methods ensure that the model is not a “black box” and can be trusted by medical professionals for decision support.

6.3 Overall Outcome

The proposed system successfully demonstrates the effectiveness of machine learning-enabled predictive analytics for early diabetes risk detection. By integrating multiple models, handling class imbalance using SMOTE, and applying robust evaluation techniques, the system achieves reliable and accurate predictions.

The final model not only provides high predictive performance but also ensures interpretability through SHAP and LIME, making it suitable for real-world healthcare applications. The ability to identify high-risk patients at an early stage can support preventive care and reduce complications associated with diabetes.

Overall, this work highlights the potential of combining advanced machine learning techniques with explainable AI to build intelligent and trustworthy healthcare systems.

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project successfully developed an intelligent machine learning-based framework for predicting the risk of Diabetes Mellitus using multiple advanced models. Various machine learning techniques including LightGBM, CatBoost, Extra Trees Classifier, and a deep learning-based Artificial Neural Network were implemented and evaluated.

The results indicate that the best-performing model (based on weighted scoring of Recall, F1 Score, and ROC-AUC) achieved superior performance in accurately identifying diabetic patients. Ensemble models demonstrated strong performance due to their ability to capture complex feature relationships, while the ANN model effectively learned nonlinear patterns in the dataset.

The integration of Explainable AI techniques such as SHAP and LIME enhanced the transparency of the system by providing both global and local interpretability. Important features like Glucose, BMI, and Insulin were identified as key contributors to diabetes prediction.

Overall, the proposed framework demonstrates the effectiveness of combining machine learning and explainable AI for early disease detection. It can assist healthcare professionals in making informed decisions and improve patient outcomes through timely diagnosis and risk assessment.

8. FUTURE SCOPE

The project can be further enhanced and extended in several ways to improve its real-world applicability and performance:

1. Real-Time Data Integration
   The system can be integrated with hospital databases or IoT-based health monitoring systems to enable real-time diabetes risk prediction.

2. Web/Mobile Application Development
   A user-friendly web or mobile application can be developed to allow patients and doctors to easily access predictions and insights.

3. Incorporation of Additional Features
   Including more parameters such as lifestyle habits, dietary patterns, physical activity, and genetic data can improve model accuracy and reliability.

4. Advanced Deep Learning Models
   More complex deep learning architectures (such as CNNs or hybrid models) can be explored for better performance on large-scale datasets.

5. Cloud Deployment and Scalability
   Deploying the system on cloud platforms can ensure scalability, remote access, and real-time monitoring capabilities.

6. Continuous Learning System
   The model can be updated periodically using new patient data to improve prediction accuracy and adapt to changing health patterns.

8. REFERENCES

The following references were used during the development of this project:

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